EXTENSION WORK

This only works for non-metals. For metals, the most reactive one is the best reducing agent. The difference comes because non-metals react to form negative ions but metals react to form positive ions.

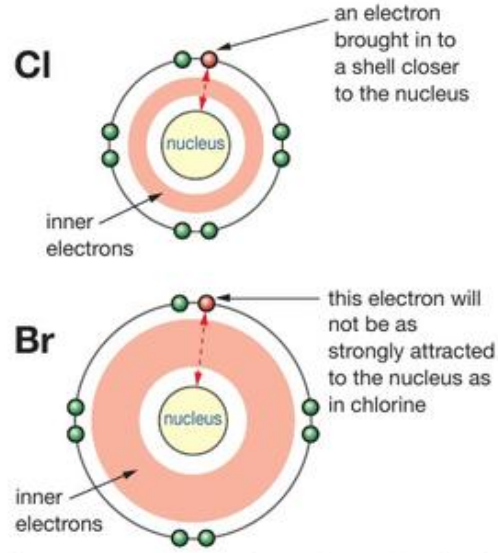
EXTENSION WORK

We can now say a bit more than just 'chlorine is more reactive than bromine, which is more reactive than iodine'. We can describe the reactivity in terms of how good an oxidising agent something is: chlorine is a stronger oxidising agent than bromine, which is a stronger oxidising agent than iodine. Chlorine can oxidise (take electrons away from) bromide ions and iodide ions; bromine can oxidise (take electrons away from) iodide ions, but not chloride ions; iodine cannot oxidise bromide ions or chloride ions.

CHEMISTRY ONLY

EXPLAINING THE TREND IN THE REACTIVITY OF THE HALOGENS

We can explain the reactivity of the halogens in terms of how readily they form negative ions. When the halogens react (in these reactions) they form 1– ions by taking electrons away from something else. A chlorine atom is smaller than a bromine atom, so when we add an electron to the outer shell of a chlorine atom, we are adding it to a shell closer to the nucleus. If it is in a shell closer to the nucleus it is more strongly attracted to the nucleus. The bromine atom is bigger and so the outer electron is added to a shell further away from the nucleus, where it is not as strongly attracted. Chlorine therefore has a stronger tendency to form a 1– ion than bromine, and a chlorine atom will take an electron away from a bromide ion.



▲ Figure 12.10 Chlorine has a stronger attraction for an electron than bromine.

EXTENSION WORK

This argument does not apply to fluorine. Fluorine atoms don't, in fact, accept electrons more easily than chlorine atoms do. This is for reasons which we don't study at International GCSE. The greater reactivity of fluorine is because of a particularly weak covalent bond between the fluorine atoms. In the exam, you should not be asked to explain why fluorine is more reactive than chlorine, but, if you are, you will have to use the explanation described here, even though it is wrong!

EXTENSION WORK

You may be thinking why doesn't the higher nuclear charge on a bromine atom mean that it attracts electrons more strongly? You have to consider the amount of attraction the incoming electron actually feels from the nucleus. In chlorine, there are 17 positively charged protons balanced by the 10 negatively charged electrons in the inner shells. That means the new electron feels an overall pull of 7+ from the centre of the atom. A similar argument with bromine shows that the new electron also feels an overall pull from the nucleus of 7+, but in the bromine case, it is further away.

This explanation will not be required at International GCSE.

END OF CHEMISTRY ONLY

SUMMARISING THE MAIN FEATURES OF THE GROUP 7 ELEMENTS

Group 7 elements:

- have diatomic molecules, X_2 (e.g. F_2, Cl_2, Br_2)
- go from gases to liquid to solid as you move down the group: the melting points and boiling points increase down the group
- have coloured poisonous vapours: the colours of the elements get darker down the group
- form compounds with the formula HX (e.g. HF, HCl, HBr) when reacted with hydrogen and these dissolve in water to form acids
- form ionic salts with metals and covalent compounds with non-metals
- form X^- (e.g. F^-, Cl^-, Br^-) ions in ionic compounds
- become less reactive towards the bottom of the group
- will displace elements lower down the group from their salts.

We can use these properties and trends to predict the properties of astatine, the element below iodine in Group 7. Astatine is an extremely rare radioactive element and no one has ever seen a sample of it, but we can predict that it:

- will be a darker colour than iodine, very dark grey or black
- will be a solid at room temperature and have a higher melting point than iodine
- will be diatomic and contain At_2 molecules
- will react with hydrogen to form HAt, which will dissolve in water to form an acid
- will form the salt NaAt with sodium or KAt with potassium
- will contain the astatide ion (At^-) in its ionic salts
- will be less reactive than iodine
- will be displaced from solution by iodine

CHAPTER QUESTIONS

SKILLS INTERPRETATION 8

1 a Draw a diagram to show the arrangement of electrons in a fluorine atom.

SKILLS REASONING 7

b Explain why fluorine is placed in Group 7 in the Periodic Table.

SKILLS INTERPRETATION 8

c Draw a dot-and-cross diagram (showing outer electrons only) of the molecule formed when fluorine reacts with hydrogen.

SKILLS REASONING 7

d Explain why fluorine has a lower melting point than chlorine.

e Explain what you would observe when a chlorine solution is added to a solution of potassium fluoride.